SOP-001: Cargo Temperature Control Limits

Document ID: SOP-001

Owner: Quality Assurance (QA) Department

Applies to: All reefer (refrigerated) trucks and containers

1. Purpose

This SOP defines the allowed temperature range for each cargo class. The IoT sensor reading (`iot_temperature`, in °C) on each vehicle must be compared against these limits.

2. Temperature Ranges by Cargo Class

Cargo class	Examples	Allowed range
Frozen goods	Ice cream, frozen meat, frozen seafood	-10°C to -5°C
Fresh perishables (chilled)	Fresh produce, dairy, fresh fish, flowers	2°C to 8°C
Pharmaceuticals and vaccines	Vaccines, insulin, biologics	2°C to 8°C (strict)
Ambient controlled goods	Chocolate, some medicines, wine	15°C to 25°C

3. Fresh Perishables Rule

- Fresh perishables must stay between **2°C and 8°C**.
- A reading **above 8°C** is a **temperature excursion**.
- A reading **below 2°C** is a **freeze risk** (produce can be damaged by freezing).
- An excursion above 8°C lasting more than **30 minutes**, or any single reading above **12°C**, is a **Tier 1 breach** (see SOP-002).
- A reading between 8°C and 12°C lasting less than 30 minutes is a **Tier 2 warning**.

4. Frozen Goods Rule

- Frozen goods must stay at or below **-5°C**.
- A reading between -5°C and 0°C is a **Tier 2 warning**.
- A reading **above 0°C** means thawing has started and is a **Tier 1 breach**.

5. Monitoring Frequency

- Sensors report at least once every hour.
- Dispatchers must review the active fleet temperature dashboard at the start of each shift and after every alert.
- If a sensor stops reporting for more than 2 hours, treat it as a **Tier 2 warning** and contact the driver to read the manual thermometer.